DysLex-Probe: Viva Preparation Report

Probing Cross-Lingual Lexical Representations in Multilingual LLMs for Low-Resource Dyslexia Screening

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Prepared as a study guide: what we did, what we achieved, what we could not do, and likely examiner questions.

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1. One-paragraph summary (say this if asked "what is your thesis about?")

Dyslexia is a phonological reading difficulty. Reliable, normed screening tests exist for English but not for Bangla, a language with over 270 million speakers. We asked whether multilingual large language models (LLMs), which are already trained on Bangla, internally carry phonological knowledge that could transfer from a high-resource language (English) to a low-resource one (Bangla). We probed four architecturally different multilingual LLMs on a purpose-built 440-item English/Bangla stimulus set. Our central result is twofold. First, a methodological warning: three of the four standard phonological probing tasks turned out to be solvable by surface shortcuts (first letter, word length), so they do not measure phonology at all. Second, on the one task that survived strict sanity controls (telling real words from pseudowords), cross-lingual transfer is genuine for the two models with real Bangla coverage (BLOOM and XLM-R) and collapses for the English-heavy or span-corruption models (Llama and mT5). The deciding factor is pretraining objective and language coverage, not model size.

2. Motivation: why this research exists

- **Clinical gap.** Dyslexia is about how the brain processes the sounds of language (phonology), not intelligence or vision. English has standardized normed screeners (e.g. CTOPP-2). Bangla has no validated automated screener; that absence is itself a barrier to early identification.
- **Scientific opportunity.** A representation that truly encodes phonology should generalize across orthographic depth. English is orthographically opaque; Bangla is intermediate (conjuncts, matra). If a model's knowledge survives that change of script, it is encoding structure, not surface statistics.
- **Practical leverage.** If one multilingual model carries transferable phonological structure, it could seed screening tools for many low-resource languages at once, instead of building one

per language.

3. Research questions

1. **RQ1 (validity):** Do the models genuinely encode phonological knowledge, or do probes succeed by exploiting surface shortcuts?
2. **RQ2 (transfer):** Does lexical knowledge learned from English transfer zero-shot to Bangla?
3. **RQ3 (localization):** Where inside a model does cross-lingual lexical knowledge live: distributed, or concentrated in specific units/layers?
4. **RQ4 (recommendation):** Given all of the above, which model is best for which purpose?

4. What we built

4.1 The stimulus dataset (440 items)

220 English + 220 Bangla, perfectly balanced, 18-column schema, Bangla items grounded in Badsha (2024) dyslexia patterns and verified by a native speaker.

Category	EN	BN	Total
Minimal pair	60	60	120
Multisyllabic	30	30	60
Pseudoword	25	25	50
Real word	25	25	50
Phoneme-grapheme	25	25	50
Rhyme	20	20	40
Sentence	15	15	30
+ extended real/pseudo sets	20	20	40
Combined total	**220**	**220**	**440**

For all honest RQ2/RQ3 reporting we used a **length-matched real-vs-pseudoword subset**: n = 50 per language, matched 1:1 on character length (mean chars EN 6.88, BN 5.20), deterministic with random_state = 42.

4.2 The four models (chosen to span the design space)

Model	Architecture	Params	Pretrain langs	Bangla coverage
XLM-R large	Encoder-only (masked LM)	0.56B	100	Explicit
BLOOM 7B1	Decoder-only (causal LM)	7.1B	46	Explicit
mT5 XL	Encoder-decoder (span corruption)	3.7B (enc 1.67B)	101	Explicit
Llama 3.1 8B	Decoder-only (causal LM)	8B	mostly EN	Minimal

The point of these four is to **separate two variables**: training objective (masked / causal / span-corruption) and language coverage (multilingual vs English-heavy), independently of raw size.

5. How we did it (methodology)

1. **Frozen representations.** Each model's weights are frozen. We pass every stimulus through and extract hidden states at each layer. We never fine-tune; we only read what is already there.
2. **Linear probing.** A simple linear classifier (logistic regression) or ridge regressor is trained on those hidden states. A linear probe is deliberately weak, so if it succeeds the information must be linearly available in the representation, not invented by the probe.
3. **Cross-lingual protocol.** Train the probe on English representations, test on Bangla. Success means knowledge transferred across the script boundary.
4. **Three sanity checks (the core methodological contribution):** a first-character lookup baseline, a 5-feature surface-statistics baseline (char length, byte length, unique chars, etc.), and length-matching of real vs pseudowords. A task only counts if the model beats these shortcuts.
5. **Causal ablation.** Using forward hooks we zero out the top-K ranked units and measure the Bangla performance drop, compared against zeroing random units. A selective drop means those specific units causally carry the cross-lingual signal.

6. What we found (results, RQ by RQ)

RQ1: most standard probing tasks are invalid

Of four tasks (voicing, real-vs-pseudoword, orthographic regularity, syllable count), **three failed the sanity checks.**

- **Voicing** labels for Bangla were 100% recoverable from the **first character alone**. A probe on

layer-0 embeddings scored F1 0.78 to 0.94 across the four models (deterministic upper bound 1.00). The probe reads letter identity, not phonology.

- **Orthographic regularity** was solved by a 5-feature surface baseline (EN F1 0.66, BN 0.82); the LLMs added at most +0.03 (EN) / +0.01 (BN).
- **Syllable count** was surface-confounded (EN surface R^2 0.81); 3 of 4 models actually underperformed the surface baseline in English.
- **Real-vs-pseudoword** was the **only survivor**, and only after length matching. Within-language Bangla peaks: BLOOM 0.94, XLM-R 0.92, mT5 0.92, Llama 0.90; every model gained between +0.20 and +0.30 in Bangla over the shortcut baselines.

RQ2: cross-lingual transfer splits the models in two

EN → BN zero-shot transfer on the length-matched real-vs-pseudoword task:

Model	Bangla F1	Gain over surface	Verdict
BLOOM 7B1	0.92	+0.30	**Real transfer**
XLM-R large	0.86	+0.24	**Real transfer**
Llama 3.1 8B	0.63	+0.01	**Collapsed to surface**
mT5 XL	0.60	−0.02	**Collapsed below surface**

The two models that transfer both have explicit Bangla coverage. Crucially, **XLM-R (0.56B) beats Llama (8B)** by a wide margin: roughly 13× smaller, far better cross-lingual transfer. **Language coverage beats scale.**

RQ3: only mT5 has localized cross-lingual coding

Causal ablation (drop in Bangla F1 from zeroing top-ranked units, minus the drop from random units = "specificity"):

Model	Specificity	Interpretation
mT5 XL	**+0.11**	Knowledge concentrated in a few units
XLM-R	+0.004	Distributed
BLOOM	−0.02	Distributed / redundant
Llama	n/a	No transfer to disrupt (BN = chance)

Repeating the ablation at five mT5 encoder depths (three seeds each) the effect held everywhere and peaked at **layer 12** (specificity +0.13). Two units, **u782 and u1880**, recurred in the top-10 across most depths, acting as a "backbone"; 31 of 38 unique top units were layer-specific. The mechanism is **hybrid**: a small persistent backbone plus layer-local specialized circuits.

RQ4: which model wins, by criterion

You want...	Use	Key number
Best Bangla transfer (prototype)	BLOOM	0.92 F1, +0.30 over surface
Best efficiency (small hardware)	XLM-R	0.86 F1 at 0.56B (~13× smaller)
Interpretability / mechanism	mT5	+0.11 specificity, backbone units
Cross-lingual Bangla work	Do NOT use Llama	collapses to 0.63

7. What we achieved (contributions)

1. **A methodological standard.** We showed 3 of 4 widely-used phonological probing tasks measure surface shortcuts, not phonology. Surface-matched stimuli plus native-verified labels are the minimum bar for clinical probing. This is the single most transferable contribution.
2. **Genuine cross-lingual lexical transfer, demonstrated and bounded.** Real-vs-pseudoword transfer to Bangla is real (+0.20 to +0.30) for multilingual models after surface controls.
3. **An empirical law for this setting:** pretraining objective + language coverage jointly decide transfer; scale alone does not.
4. **An interpretability result:** mT5 uniquely localizes cross-lingual lexical coding into a few units with a backbone-plus-routing structure.
5. **A reusable, balanced, native-verified 440-item EN/BN stimulus dataset** with an 18-column schema.

8. What we could NOT do (limitations: state these proactively)

- **Not a clinical tool.** 0.92 F1 on a research task is not a diagnosis. Using it to label a child would be harmful. This is a representational probe, not a screener yet.

- **Small evaluation sets.** The honest subset is n = 50 matched pairs per language, so per-layer scores are point estimates with real variance.
- **Architectural confound for the mT5 finding.** mT5 was the only encoder-decoder tested, so its localization could be due to the span-corruption objective, the encoder-decoder split, or both. We cannot separate these without more encoder-decoder models.
- **Ablation baseline ceiling.** After matching, the frozen-probe English baseline is ~1.00, so only the Bangla-side drop is an interpretable causal signal.
- **BLOOM ablation specificity slightly negative (−0.02):** ablation marginally improved Bangla transfer, indicating distributed redundancy rather than a clean null.
- **Unusable original labels.** The Bangla voicing column is inferable from the first character and had to be discarded for phonological probing.
- **Coverage gaps.** Standard Bangla forms only; dialectal variation and conjunct-encoding variants not modelled. No real dyslexic-child data used.

9. Numbers cheat-sheet (memorize these)

- Dataset: **440** items (220 EN + 220 BN), 7 categories, 18 columns. Honest subset n = **50**/language, random_state 42.
- Voicing shortcut: first-character probe F1 **0.78 to 0.94**; 100% of BN voicing labels first-char recoverable.
- Real-vs-pseudoword in-language BN peaks: BLOOM **0.94**, XLM-R **0.92**, mT5 **0.92**, Llama **0.90**; gains **+0.20 to +0.30**.
- Cross-lingual transfer (EN→BN): BLOOM **0.92** (+0.30), XLM-R **0.86** (+0.24), Llama **0.63** (+0.01), mT5 **0.60** (−0.02).
- mT5 ablation specificity **+0.11**, peak at layer **12** (+0.13); backbone units **u782, u1880**.
- Efficiency headline: XLM-R 0.56B beats Llama 8B (~**13×** smaller) on transfer.

10. Glossary (be ready to define on the spot)

- **Phonology:** the sound structure of language; dyslexia is primarily a phonological-processing deficit.
- **Orthographic depth:** how directly spelling maps to sound. English is opaque; Bangla is intermediate.
- **Linear probe:** a simple linear classifier trained on frozen hidden states to test what information is linearly readable.
- **Frozen model:** weights unchanged; we read existing representations rather than retraining.
- **Causal ablation:** deliberately zeroing units to test whether they causally carry a signal (vs merely correlating).

- **Specificity (our metric):** drop from ablating top-ranked units minus drop from ablating random units.
- **F1 score:** the harmonic mean of precision and recall; used because classes are balanced and we want both error types penalized.
- **Pseudoword:** a pronounceable non-word obeying the language's phonotactics (e.g. "blicket").
- **Zero-shot transfer:** train in one language, test in another with no target-language training.
- **Span corruption:** mT5's pretraining objective (mask and reconstruct spans), distinct from masked-token (XLM-R) or next-token (BLOOM/Llama) objectives.

11. Anticipated viva questions with model answers

Q1. Why these four specific models?

They span the two variables that matter: pretraining objective (masked encoder = XLM-R, causal decoder = BLOOM and Llama, encoder-decoder = mT5) and language coverage (multilingual-with-Bangla vs English-heavy = Llama). Holding the task fixed and varying these lets us attribute differences to design choices rather than to one arbitrary model.

Q2. Why probe a frozen model instead of fine-tuning?

We want to know what knowledge the pretraining already produced. Fine-tuning would inject new task-specific knowledge and confound the question. A frozen model plus a weak linear probe isolates pre-existing, linearly-accessible structure.

Q3. Why is the first-character voicing finding such a big deal?

Because it invalidates an entire class of "the model understands phonology" claims. If a layer-0 embedding probe scores up to 0.94 using only the first letter, then high probe accuracy proves nothing about phonological processing. It forced us to discard the voicing task and re-examine the others, and three of four failed. This is why we frame the methodological lesson as our main contribution.

Q4. mT5 collapses on transfer (0.60) yet you call it the "interpretable winner". Isn't that contradictory?

No, they are different axes. Raw transfer asks "how well does English knowledge work on Bangla?" mT5 is weak there. Localization asks "where does whatever cross-lingual signal exists live?" mT5 is the only model where ablating a few units selectively destroys the Bangla signal (specificity +0.11), so its representation is recoverable and inspectable. A model can encode a modest signal in a clean, localized way; that is scientifically valuable even if the signal is small.

Q5. Why does a 0.56B model (XLM-R) beat an 8B model (Llama)?

Because Llama saw almost no Bangla during pretraining, so it has little Bangla lexical structure to

transfer, regardless of size. XLM-R is a multilingual masked encoder with explicit Bangla coverage. This is the headline: coverage and objective beat scale for low-resource transfer.

Q6. How do you know the probe is not just memorizing the training words?

Three guards: the probe is linear (low capacity), we test cross-lingually on a different script, and we compare against surface and first-character baselines plus length-matching. Beating all of those is hard to achieve by memorization.

Q7. Is n = 50 per language enough to trust?

It is small, and we say so explicitly. We treat per-layer numbers as point estimates, not guarantees, and we lean on effects that are large (gains of +0.20 to +0.30) and consistent across all four models and across layers/seeds, rather than on any single fragile number.

Q8. What exactly does "ablation specificity" mean, and why compare to random units?

Specificity = (Bangla drop when we zero the top-ranked units) minus (Bangla drop when we zero random units). The random comparison controls for the trivial damage any ablation causes. A positive specificity means the top units matter more than chance, i.e. the signal is concentrated there. Only mT5 shows a clear positive value (+0.11).

Q9. Could the mT5 localization just be an artifact of it being your only encoder-decoder?

Yes, that is an honest confound we state. The effect could come from the encoder-decoder split, the span-corruption objective, or both. We mitigate the "fluke" worry by showing it persists across five depths and three seeds with a stable backbone (u782, u1880), but disentangling architecture from objective needs more encoder-decoder models, which we list as future work.

Q10. Why F1 rather than plain accuracy?

F1 balances precision and recall and is robust when we care about both false positives and false negatives. With balanced classes it is a fair, single-number summary and is the standard for this kind of probing.

Q11. Why English and Bangla specifically?

They differ in orthographic depth: English is opaque, Bangla is intermediate with conjuncts and matra. A signal that survives that script change is structural, not surface. And Bangla is a high-impact low-resource case with no existing screener.

Q12. Is this a dyslexia diagnostic tool?

No. It is a representational probe. It tells us whether models carry transferable lexical/phonological

structure. Turning that into a screener requires clinical validation with real children and experts, which is explicitly future work. We are careful never to overclaim.

Q13. What is the real-world significance if it is not a tool yet?

Two things. First, the methodology stops the field from reporting false positives in cross-lingual phonological probing. Second, it gives a concrete model-selection answer for anyone building low-resource screening prototypes: start from BLOOM or XLM-R, not from a bigger English-centric model.

Q14. What would you do differently or next?

Add more encoder-decoder models to resolve the mT5 confound, enlarge and diversify the Bangla data (dialects, more conjuncts), collect real dyslexic-reader data, and design future labels that cannot be shortcut. We would also report confidence intervals given the small n.

Q15. What is your single most important finding?

That three of four standard phonological probing tasks measure surface shortcuts rather than phonology, and that once you fix this, multilingual coverage (not scale) is what makes cross-lingual lexical transfer to Bangla work.

> **Division of labour tip for the panel:** agree in advance who answers methodology questions (probing, ablation, sanity checks), who answers results/numbers, who answers clinical/impact/limitations, and who answers dataset questions. If a question lands outside your slides, hand off cleanly: "My colleague worked on that part, [name] will take it."

All figures in this report are taken directly from the project's own results files (comparison_four_models.md and the faculty findings report). No numbers were invented or altered.